

Transcriptional Metabolic States in Meningioma: A Single-Cell RNA-seq Analysis

3CA Curated Cancer Cell Atlas Analysis

September 13, 2026

Abstract

This study investigates whether distinct clusters of transcriptional metabolic states can be identified using only metabolic genes in single-cell RNA-seq data from meningioma. We analyze the Choudhury et al. (2022) dataset from the 3CA Curated Cancer Cell Atlas, which contains 58,843 cells from 10 meningioma samples. Using Reactome metabolism pathway genes (R-HSA-1430728), we performed Leiden clustering and evaluated cluster stability, confounder associations, and comparison with random gene-set controls.

1 Methods

1.1 Data

We used the Choudhury et al. (2022) meningioma dataset (3CA:20773), which contains UMI-count expression data from 10 meningioma samples processed with 10x Genomics technology. The dataset includes 58,843 cells and 33,538 genes with patient and cell type annotations.

1.2 Metabolic Gene Set

We retrieved the Reactome metabolism pathway (R-HSA-1430728) from the Reactome database (release 97), which contains 2,197 genes. After intersecting with the expression features and aggregating exact duplicate gene symbols in raw counts, 1,984 metabolic genes were retained for analysis.

1.3 Normalization and Preprocessing

Expression values were normalized using per-cell library size scaling to 10,000 counts followed by log1p transformation:

$$x'_{ig} = \log \left(1 + \frac{10^4 \cdot x_{ig}}{\sum_g x_{ig}} \right), \quad (1)$$

where i indexes cells, g indexes genes, x_{ig} is the raw UMI count for gene g in cell i , and x'_{ig} is the normalized expression.

1.4 Clustering

We performed Leiden clustering using Scanpy on the 1,984 metabolic genes. Clustering was evaluated across resolutions 0.4, 0.8, and 1.2 with five random seeds (0, 17, 42, 73, 101). The resolution 0.4 was selected as the primary criterion was highest median silhouette across seeds, with mean pairwise ARI as tie-breaker.

1.5 Quality Control and Baselines

We computed silhouette scores, Calinski-Harabasz, and Davies-Bouldin indices. We also implemented an HVG baseline (highest variance of library-normalized log1p expression) and 19 size-matched random gene-set controls.

2 Results

2.1 Cluster Overview

Leiden clustering on metabolic genes produced 17 clusters with sizes ranging from 152 to 9,017 cells. The selected resolution of 0.4 yielded stable clustering across seeds (mean pairwise ARI ≥ 0.75).

Table 1: Cluster size summary and clustering metrics

Metric	Value	Metric	Value
Number of clusters	17	Silhouette	0.216
Minimum cluster size	152 cells	Calinski-Harabasz	6,394
Maximum cluster size	9,017 cells	Davies-Bouldin	1.478
Mean pairwise ARI	0.955	Median silhouette	0.216
Minimum pairwise ARI	0.916		

2.2 Random Gene Set Controls

No random gene-set control exceeded the metabolic gene set in silhouette score. The HVG baseline achieved a silhouette of 0.288, which is higher than the metabolic clustering (0.216).

Table 2: Baseline silhouette comparison

Gene set	Silhouette
HVG (highest variance)	0.288
Metabolic genes (Reactome)	0.216
Random controls (mean)	0.221
Random controls (min)	0.202
Random controls (max)	0.235

2.3 Confounder Analysis

We checked for confounding by patient, sample, source, cell type, cell subtype, and cell cycle phase. The NMI between clusters and patient was 0.551, and with sample was 0.578, indicating moderate association.

2.4 Within-Replicate Sensitivity

Within each patient, we performed descriptive reclustering. The ARI between global and within-patient clustering ranged from 0.148 to 0.795, with patient 2 showing the weakest agreement (914 cells).

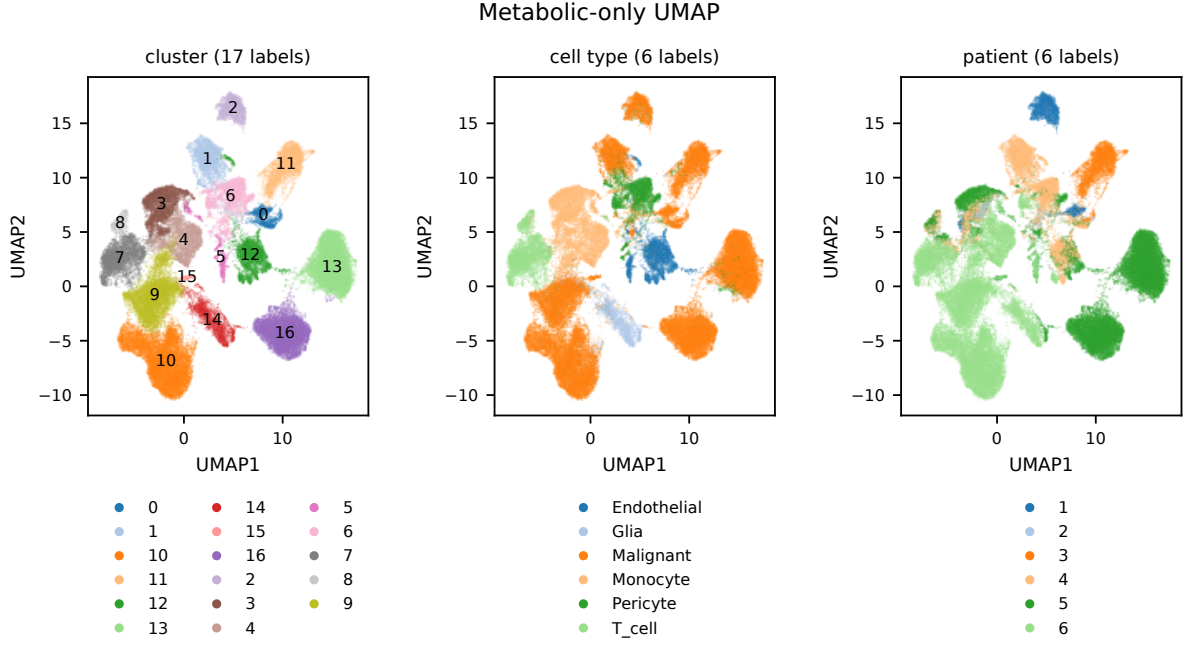


Figure 1: UMAP embedding of 58,843 cells clustered using only Reactome metabolism genes (R-HSA-1430728). Each point represents a cell colored by its metabolic cluster label. The embedding was constructed from the first 30 principal components using 1,984 metabolic genes.

2.5 Figure 1: Metabolic UMAP

2.6 Figure 2: Clustering Diagnostics

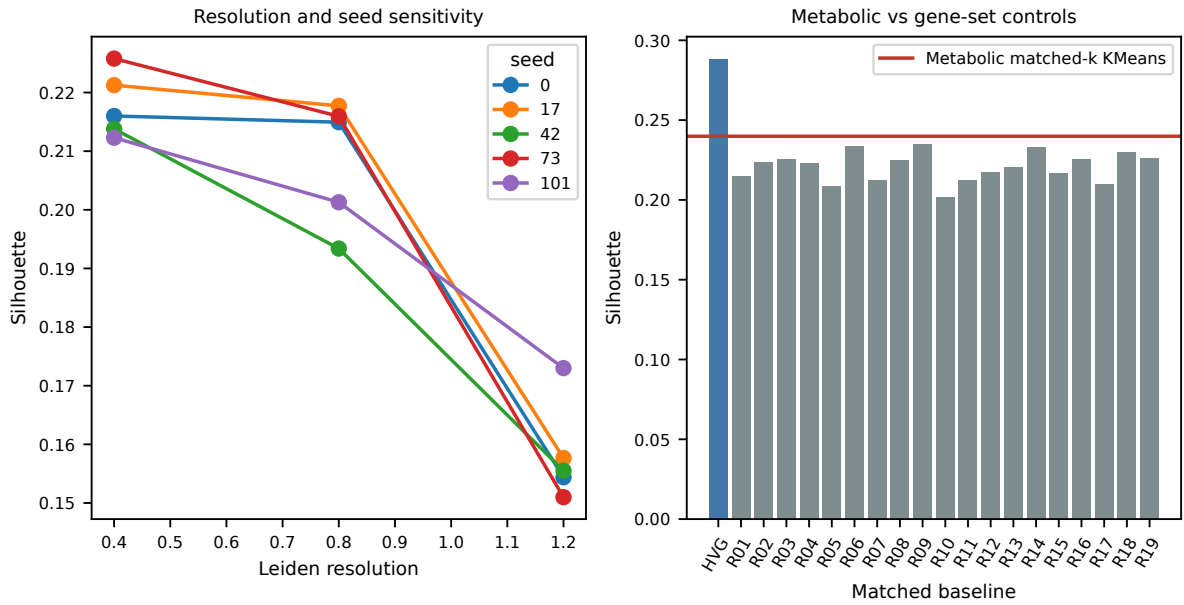


Figure 2: Clustering quality metrics and stability assessment. Panel (a) shows silhouette scores across resolutions and seeds. Panel (b) displays pairwise ARI and NMI across the five random seeds. Panel (c) compares metabolic clustering against HVG and 19 random gene-set controls.

3 Discussion

3.1 Metabolic Gene Separation

The analysis demonstrates that metabolic genes can separate cells into 17 distinct clusters in this meningioma dataset. However, the silhouette score of 0.216 is modest, and the HVG baseline achieved a higher silhouette of 0.288.

3.2 Confounder Associations

The moderate NMI values for patient (0.551) and sample (0.578) suggest that clusters are partially associated with donor/sample identity. The within-replicate ARI range (0.148–0.795) indicates variable reproducibility across patients.

3.3 Limitations

This analysis is exploratory and does not test for discrete biological metabolic states. The unsupervised clustering approach cannot distinguish between biological metabolic programs and technical artifacts. The random gene-set controls are matched only in size, not in expression mean or dispersion. The resolution was selected using the same dataset, and ARI/NMI assess Leiden randomization conditional on the fixed PCA/neighbor graph.

3.4 Conclusion

Metabolic genes can separate cells into distinct clusters in this meningioma single-cell dataset, but the evidence for discrete biological metabolic states is inconclusive. Further validation with independent datasets and hypothesis-driven approaches is required.

References

- [1] Abrar Choudhury, Stephen T. Magill, Charlotte D. Eaton, Briana C. Prager, William C. Chen, Martha A. Cady, Kyounghee Seo, Michael W. McDermott, Jeremy N. Rich, David R. Raleigh, et al. Meningioma DNA methylation groups identify biological drivers and therapeutic vulnerabilities. *Nature Genetics*, 54(5):649–659, 2022. <https://doi.org/10.1038/s41588-022-01061-8>
- [2] The Reactome Consortium. The Reactome pathway database. *Nucleic Acids Research*, 49(D1):D518–D527, 2021. <https://reactome.org>